

Policy Gradient Methods

Reinforcement Learning

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Outline

Background

- Monte Carlo methods
- Importance weighting
- Control variates
- “Log-derivative trick”

Policy gradient methods

- Performance difference lemma
- Policy gradient for undiscounted episodic tasks

Policy gradients in the discounted case

- Improper state distribution
- Discounted performance difference lemma
- Policy gradient for discounted episodic tasks

REINFORCE and policy gradients in the wild

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Warmup: Monte Carlo methods

Crude Monte Carlo methods approximate an integral

$$\mathbb{E}_{z \sim p}[f(z)] = \int f(z)p(z)\mathrm{d}z$$

by

$$\frac{1}{n} \sum_{i=1}^n f(z_i) \xrightarrow{n \rightarrow \infty} \mathbb{E}_{z \sim p}[f(z)] \quad ,$$

where

$$z_i \sim p$$

for $i = 1, \dots, n$.

Warmup: Importance sampling

For two distributions p and q with $p(z) > 0 \Rightarrow q(z) > 0$ we have

$$\mathbb{E}_{z \sim p}[f(z)] = \int \frac{p(z)}{q(z)} f(z) q(z) \mathrm{d}z = \mathbb{E}_{z \sim q} \left[\frac{p(z)}{q(z)} f(z) \right]$$

and thus

$$\frac{1}{n} \sum_{i=1}^n \frac{p(z_i)}{q(z_i)} f(z_i) \xrightarrow{n \rightarrow \infty} \mathbb{E}_{z \sim p}[f(z)] \quad ,$$

where $z_i \sim q$ for $i = 1, \dots, n$, that is, we approximate the expectation over p using samples from q .

Warmup: Control variates

Assume we want to estimate $\mathbb{E}[f]$ via Monte Carlo and are worried about the variance of our estimator.

We introduce a helper function (control variate) ϕ correlated with f for which we know $\bar{\phi} = \mathbb{E}[\phi]$ and write

$$\mathbb{E}[f] = \mathbb{E}[f - \phi] + \mathbb{E}[\phi] = \mathbb{E}[f - \phi] + \bar{\phi} = \mathbb{E}[f - \phi + \bar{\phi}] .$$

We have

$$\text{Var}[f - \phi + \bar{\phi}] = \text{Var}[f - \phi] = \text{Var}[f] - 2 \text{Cov}[f, \phi] + \text{Var}[\phi] .$$

Thus, the control variate reduces the variance in the sense that $\text{Var}[f - \phi] < \text{Var}[f]$ if $2 \text{Cov}[f, \phi] > \text{Var}[\phi]$.

Analogously, $\text{Var}[f + \phi] < \text{Var}[f]$ if $2 \text{Cov}[f, \phi] < \text{Var}[\phi]$.

Warmup: “Log-derivative trick”

We have

$$\nabla_{\theta} \log p_{\theta}(z) = \frac{\nabla_{\theta} p_{\theta}(z)}{p_{\theta}(z)} \Rightarrow \nabla_{\theta} p_{\theta}(z) = p_{\theta}(z) \nabla_{\theta} \log p_{\theta}(z)$$

which implies

$$\nabla_{\theta} \mathbb{E}[f(z)] = \int f(z) \nabla_{\theta} p_{\theta}(z) \mathrm{d}z = \mathbb{E}[f(z) \nabla_{\theta} \log p_{\theta}(z)] \quad ,$$

where the RHS lends itself to Monte Carlo estimation via

$$\frac{1}{n} \sum_{i=1}^n f(z_i) \nabla_{\theta} \log p_{\theta}(z_i)$$

with $z_i \sim p$ for $i = 1, \dots, n$. $\nabla_{\theta} \log p_{\theta}(z)$ is called *score function*.

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Notation

- Discrete time t , states S , actions A
- State-action-reward sequence $s_0, a_0, r_1, \dots$, with first time arriving in a terminal state in episodic tasks denoted by T (and for $t > T$: $r_t = 0$ and $s_t = s_T$), start state distribution p_{start}
- $P_{ss'}^a = \Pr\{s_{t+1} = s' \mid s_t = s, a_t = a\}$ and $R_{ss'}^a = \mathbb{E}\{r_{t+1} \mid s_t = s, s_{t+1} = s', a_t = a\}$
- $R_s^a = \mathbb{E}\{r_{t+1} \mid s_t = s, a_t = a\} = \sum_{s'} P_{ss'}^a R_{ss'}^a$
- $\pi_{\theta}(s, a) = \pi(s, a) = \Pr\{a_t = a \mid s_t = s, \theta\}$
- $\Pr\{s \xrightarrow{k} x \mid \pi\}$: probability of going from state s to state x in k steps under policy π ($\Pr\{s \xrightarrow{0} s \mid \pi\} = 1$ and $\Pr\{s \xrightarrow{0} s' \mid \pi\} = 0$ for $s \neq s'$)

Stochastic vs. deterministic policy

- Toy Markov Decision Process (MDP) with $S = \{s, s_T\}$, s_T is terminal state, s is start state, $\gamma = 1$, $\mathcal{A} = \{\text{left}, \text{right}\}$, all actions lead to s_T , $R_s^{\text{left}} = 1$, $R_s^{\text{right}} = 0$
- Deterministic policy:

$$\pi_{\theta}(s) = \begin{cases} \text{left} & \text{if } \theta \geq 0 \\ \text{right} & \text{otherwise} \end{cases}$$

- Stochastic policy:

$$\pi'_{\theta}(s) = \begin{cases} \text{left} & \text{with probability } \sigma(\theta) \\ \text{right} & \text{with probability } 1 - \sigma(\theta) \end{cases} \quad \text{with} \quad \sigma(\theta) = \frac{1}{1 + e^{-\theta}}$$

- $V^{\pi_{\theta}}(s) = \mathbb{I}[\theta \geq 0]$, $V^{\pi'_{\theta}}(s) = \sigma(\theta)$
- $\frac{\partial}{\partial \theta} V^{\pi'_{\theta}}(s) = (1 - \sigma(\theta))\sigma(\theta)$
- $\frac{\partial}{\partial \theta} V^{\pi_{\theta}}(s) = ?$

Introduction: Value function approaches to RL

- “Standard approach” to reinforcement learning (RL) is to
 - estimate a value function (V - or Q -function) and then
 - define a “greedy” policy on top of it.
- One may argue that this approach is somehow “indirect”.
- “Strong causality” violated (small changes may have drastic effects)

Introduction: Policy gradient approaches to RL

- Model a stochastic policy by a function approximator (the “actor”) with own parameters θ , for example for discrete action set

$$\pi_{\theta}(s, a) = \frac{e^{h_{\theta}(s, a)}}{\sum_{a'} e^{h_{\theta}(s, a')}}$$

with *preferences* $h_{\theta}(s, a)$

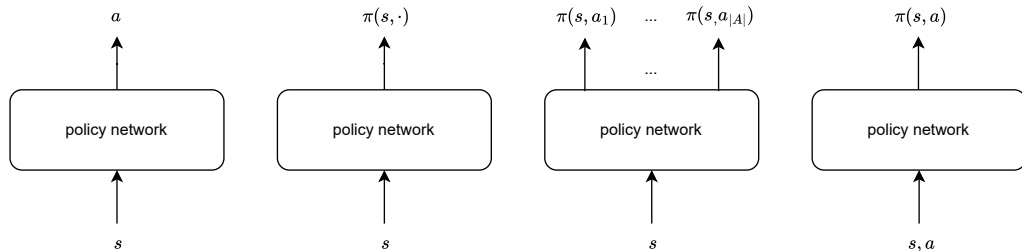
- Adapt policy according to

$$\Delta\theta \approx \alpha \nabla_{\theta} J(\pi_{\theta})$$

where $J(\pi_{\theta})$ is a performance measure of the policy π_{θ} and α a positive step-size/learning-rate

- Subsumes known methods such as actor-critic approaches and the REINFORCE algorithms

Policy networks



Example: Softmax policy

Consider vector of features ($\rightarrow$ neural network) $\phi(s, a), \forall a \in A, s \in S$; policy is a Gibbs distribution in a linear combination of the features

$$\pi(s, a) = \frac{e^{\boldsymbol{\theta}^\top \phi(s, a)}}{\sum_b e^{\boldsymbol{\theta}^\top \phi(s, b)}} .$$

Alternatively, for discrete actions, consider a feature map ϕ for the features and

$$\pi(s, a) = \frac{e^{\boldsymbol{\theta}_a^\top \phi(s)}}{\sum_b e^{\boldsymbol{\theta}_b^\top \phi(s)}}$$

with parameters $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_{|A|})$.

Start-state formulation

Goal is to maximize the expected return

$$J(\pi) = \mathbb{E}^{\pi}[R] = \mathbb{E}^{\pi} \left[\sum_{t=1}^{\infty} \gamma^{t-1} r_t \right] = \mathbb{E}_{s_0 \sim p_{\text{start}}} [V^{\pi}(s_0)]$$

with $\gamma \in [0, 1]$, $\gamma = 1$ only for episodic tasks, and

$$Q^{\pi}(s, a) = \mathbb{E}^{\pi} \left[\sum_{k=1}^{\infty} \gamma^{k-1} r_{t+k} \mid s_t = s, a_t = a, \right] .$$

Times spent in a state

- In episodic tasks, we assume a set of goal states $S_{\text{goal}} \subset S$, which are not left and any action in these states yields zero reward. An episode ends when such a state is reached, which we assume to happen with probability one in finite time.
- Let $\eta^\pi(s)$ denote the number of time steps spent, on average, in $s \in S \setminus S_{\text{goal}}$ in a single episode:

$$\eta^\pi(s) = \mathbb{E}_{s_0 \sim p_{\text{start}}} \left[\sum_{k=0}^{\infty} \Pr\{s_0 \xrightarrow{k} s \mid \pi\} \right]$$

- Time is spent in a state s if episodes start in s , or if transitions are made into s from a preceding state s' in which time is spent:

$$\begin{aligned} \eta^\pi(s) &= p_{\text{start}}(s) + \sum_{s' \in S} \eta^\pi(s') \sum_{a \in A} P_{s's}^a \pi(s', a) \\ &= p_{\text{start}}(s) + \sum_{s' \in S \setminus S_{\text{goal}}} \eta^\pi(s') \sum_{a \in A} P_{s's}^a \pi(s', a) \end{aligned}$$

Times spent in a state

- Let $S \setminus S_{\text{goal}} = \{1, \dots, \bar{n}\}$ and $\bar{\mathbf{p}}_{\text{start}}, \boldsymbol{\eta}^{\pi} \in \mathbb{R}^{\bar{n}}$ have components $[\bar{\mathbf{p}}_{\text{start}}]_i = p_{\text{start}}(i)$ and $[\boldsymbol{\eta}^{\pi}]_i = \eta^{\pi}(i)$.
- Terminal state is reached in a finite number of steps with probability one $\Rightarrow \boldsymbol{\eta}^{\pi}$ is uniquely defined with finite components with probability one.
- Let $\bar{\mathbf{P}}^{\pi} \in \mathbb{R}^{\bar{n} \times \bar{n}}$ be the matrix with components $[\mathbf{P}^{\pi}]_{i,j} = \sum_{a \in A} P_{ij}^a \pi(i, a)$, giving

$$\boldsymbol{\eta}^{\pi} = \bar{\mathbf{p}}_{\text{start}} + [\bar{\mathbf{P}}^{\pi}]^{\top} \boldsymbol{\eta}^{\pi} .$$

- Because there is a unique solution for $\boldsymbol{\eta}^{\pi}$, $(\mathbf{I} - [\bar{\mathbf{P}}^{\pi}]^{\top})^{-1} \bar{\mathbf{p}}_{\text{start}}$ must have a unique solution and $(\mathbf{I} - [\bar{\mathbf{P}}^{\pi}])^{-1}$ is invertible:

$$\boldsymbol{\eta}^{\pi} = (\mathbf{I} - [\bar{\mathbf{P}}^{\pi}]^{\top})^{-1} \bar{\mathbf{p}}_{\text{start}} = [(\mathbf{I} - \bar{\mathbf{P}}^{\pi})^{-1}]^{\top} \bar{\mathbf{p}}_{\text{start}}$$

Performance difference lemma

Lemma (performance difference lemma)

For any two policies π and π' we have

$$J(\pi) = J(\pi_{\text{ref}}) + \sum_s \eta^\pi(s) \sum_a \pi(s, a) (Q^{\pi_{\text{ref}}}(s, a) - V^{\pi_{\text{ref}}}(s))$$

Proof performance difference lemma I

Any action in a terminal state leads to zero reward and no change in state:

$$V^{\pi_{\text{ref}}}(s) = \begin{cases} 0 & \text{if } s \in S_{\text{goal}} \\ \sum_{a \in A} \pi_{\text{ref}}(s, a) \sum_{s' \in S \setminus S_{\text{goal}}} P_{s,s'}^a (R_{ss'}^a + V^{\pi_{\text{ref}}}(s')) & \text{otherwise} \end{cases} .$$

Let the non-goal states be $S \setminus S_{\text{goal}} = \{1, \dots, \bar{n}\}$.

We have

$$\bar{\mathbf{v}}^{\pi_{\text{ref}}} = (\mathbf{I} - \bar{\mathbf{P}}^{\pi_{\text{ref}}})^{-1} \bar{\mathbf{r}}^{\pi_{\text{ref}}} ,$$

where

$$\bar{\mathbf{r}}^{\pi_{\text{ref}}} = \sum_{a \in A} \sum_{j \in S} P_{ij}^a \pi_{\text{ref}}(i, a) R_{ij}^a$$

for $i \in S \setminus S_{\text{goal}}$.

Proof performance difference lemma II

We have $J(\pi_{\text{ref}}) = [\bar{\mathbf{p}}_{\text{start}}]^\top \bar{\mathbf{v}}^{\pi_{\text{ref}}}$ and $[\boldsymbol{\eta}^{\pi'}]^\top = [\bar{\mathbf{p}}_{\text{start}}]^\top (\mathbf{I} - \bar{\mathbf{P}}^\pi)^{-1}$ and $\bar{\mathbf{v}}^{\pi_{\text{ref}}} = (\mathbf{I} - \bar{\mathbf{P}}^{\pi_{\text{ref}}})^{-1} \bar{\mathbf{r}}^{\pi_{\text{ref}}}$, giving

$$\begin{aligned} \bar{\mathbf{v}}^\pi - \bar{\mathbf{v}}^{\pi_{\text{ref}}} &= (\mathbf{I} - \bar{\mathbf{P}}^\pi)^{-1} \bar{\mathbf{r}}^\pi - \overbrace{(\mathbf{I} - \bar{\mathbf{P}}^\pi)^{-1} (\mathbf{I} - \bar{\mathbf{P}}^\pi)}^{\mathbf{I}} \bar{\mathbf{v}}^{\pi_{\text{ref}}} \\ &= (\mathbf{I} - \bar{\mathbf{P}}^\pi)^{-1} (\bar{\mathbf{r}}^\pi + \bar{\mathbf{P}}^\pi \bar{\mathbf{v}}^{\pi_{\text{ref}}} - \bar{\mathbf{v}}^{\pi_{\text{ref}}}) . \end{aligned}$$

Hence, multiplying both sides from the left by $[\bar{\mathbf{p}}_{\text{start}}]^\top$ gives

$$\begin{aligned} J(\pi) - J(\pi_{\text{ref}}) &= [\boldsymbol{\eta}^{\pi'}]^\top (\bar{\mathbf{r}}^{\pi_{\text{ref}}} + \bar{\mathbf{P}}^\pi \bar{\mathbf{v}}^{\pi_{\text{ref}}} - \bar{\mathbf{v}}^{\pi_{\text{ref}}}) \\ &= \sum_s \eta^\pi(s) \sum_a \pi(s, a) (Q^{\pi_{\text{ref}}}(s, a) - V^{\pi_{\text{ref}}}(s)) . \end{aligned}$$

Policy gradient for undiscounted episodic tasks

Theorem (policy gradient for undiscounted episodic tasks)

For any differentiable stochastic policy π_{θ} and any undiscounted MDP in the start-state setting, it holds

$$\nabla_{\theta} J(\pi_{\theta}) = \sum_s \eta^{\pi_{\theta}}(s) \sum_a \nabla_{\theta} \pi(s, a) Q^{\pi_{\theta}}(s, a) \ .$$

Exciting: No $\nabla_{\theta} \eta^{\pi}$ terms!

Note: Action a is selected uniformly

Proof policy gradient for undiscounted tasks

Consider the i th component of the gradient, and let e_i be the i th unit vector. We have

$$\begin{aligned}\frac{\partial}{\partial \theta_i} J(\pi_\theta) &= \lim_{\epsilon \rightarrow 0} \frac{J(\pi_{\theta + \epsilon e_i}) - J(\pi)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{\sum_s \eta^{\pi_{\theta + \epsilon e_i}} \sum_a \pi_{\theta + \epsilon e_i}(s, a) (Q^{\pi_\theta}(s, a) - V^{\pi_\theta}(s))}{\epsilon}\end{aligned}$$

and because $\sum_a \pi_{\theta + \epsilon e_i}(s, a) V^{\pi_\theta}(s) = V^{\pi_\theta}(s) = \sum_a \pi_\theta(s, a) Q^{\pi_\theta}(s, a)$

$$\begin{aligned}&= \lim_{\epsilon \rightarrow 0} \frac{\sum_s \eta^{\pi_{\theta + \epsilon e_i}} \sum_a (\pi_{\theta + \epsilon e_i}(s, a) - \pi_\theta(s, a)) Q^{\pi_\theta}(s, a)}{\epsilon} \\ &= \sum_s \eta^{\pi_\theta}(s) \sum_a \frac{\partial \pi_\theta(s, a)}{\partial \theta_i} Q^{\pi_\theta}(s, a) \quad .\end{aligned}$$

As this holds for all i , we get the expression for $\nabla_\theta J(\pi_\theta)$.

Policy gradient theorem, MC friendly

Corollary

For any differentiable stochastic policy π_{θ} and any undiscounted MDP in the start-state setting it holds

$$\nabla_{\theta} J(\pi_{\theta}) \propto \mathbb{E}^{\pi} \left[\frac{1}{\pi_{\theta}(s, a)} \nabla_{\theta} \pi(s, a) Q^{\pi_{\theta}}(s, a) \right] .$$

Great for Monte Carlo approximation:

$$\nabla_{\theta} J(\pi_{\theta}) \propto \mathbb{E}^{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s, a) Q^{\pi_{\theta}}(s, a)]$$

$\frac{1}{\pi_{\theta}(s, a)}$ accounts for sampling of actions according to π instead of uniformly

Deterministic policy gradient theorem, undiscounted

Theorem (deterministic policy gradient for undiscounted episodic tasks)

For any MDP in the start-state setting and any deterministic policy π_{θ} it holds

$$\nabla_{\theta} J(\pi_{\theta}) = \sum_s \eta^{\pi_{\theta}}(s) \nabla_{\theta} \pi_{\theta}(s) \nabla_{\theta} Q^{\pi_{\theta}}(s, a)|_{a=\pi_{\theta}(s)} .$$

By the chain rule we have

$$\nabla_{\theta} Q^{\pi_{\theta}}(s, \pi_{\theta}(s)) \stackrel{\text{chain rule}}{=} \nabla_a Q^{\pi_{\theta}}(s, a)|_{a=\pi_{\theta}(s)} \nabla_{\theta} \pi_{\theta}(s)$$

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Improper state distribution

Discounted state (or improper) distribution:

$$\begin{aligned}\eta_{\gamma}^{\pi_{\theta}}(s) &= \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k \Pr\{s_0 \xrightarrow{k} s \mid \pi_{\theta}\} \right] \\ &= \sum_{t=0}^{\infty} \gamma^t \Pr\{s_t = s \mid \pi_{\theta}\} .\end{aligned}$$

- The expectation is over all trajectories following π_{θ} and $s_0 \sim p_{\text{start}}$.
- $\eta_{\gamma}^{\pi_{\theta}}$ is not normalized ($\sum_s \eta_{\gamma}^{\pi_{\theta}}(s) \neq 1$ for $\gamma < 1$), a factor $(1 - \gamma)$ is missing.

This follows from $\sum_s \Pr\{s_0 \xrightarrow{k} s \mid \pi_{\theta}\} = 1$ for all k (you have to be in some state after k steps) and from the geometric series $\sum_{k=0}^{\infty} \gamma^k = (1 - \gamma)^{-1}$.

Performance difference lemma

Lemma (performance difference lemma, discounted case)

For any two discounted policies π_{ref} and π we have:

$$\begin{aligned} J(\pi) &= J(\pi_{\text{ref}}) + \mathbb{E}^{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (Q^{\pi_{\text{ref}}}(s_t, a_t) - V^{\pi_{\text{ref}}}(s_t)) \right] \\ &= J(\pi_{\text{ref}}) + \sum_s \eta_{\gamma}^{\pi}(s) \sum_a \pi(s, a) (Q^{\pi_{\text{ref}}}(s, a) - V^{\pi_{\text{ref}}}(s)) \end{aligned}$$

Proof discounted performance difference lemma, 1st equality

$$\begin{aligned}\mathbb{E}^{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (Q^{\pi_{\text{ref}}}(s_t, a_t) - V^{\pi_{\text{ref}}}(s_t)) \right] \\&= \mathbb{E}^{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (r_{t+1} + \gamma V^{\pi_{\text{ref}}}(s_{t+1}) - V^{\pi_{\text{ref}}}(s_t)) \right] \\&= \mathbb{E}^{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} + \sum_{t=0}^{\infty} (\gamma^{t+1} V^{\pi_{\text{ref}}}(s_{t+1}) - \gamma^t V^{\pi_{\text{ref}}}(s_t)) \right] \\&= \mathbb{E}^{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} - V^{\pi_{\text{ref}}}(s_0) \right] = J(\pi) - J(\pi_{\text{ref}})\end{aligned}$$

Proof discounted performance difference lemma, 2nd equality

$$\begin{aligned} J(\pi) &= J(\pi_{\text{ref}}) + \mathbb{E}^{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (Q^{\pi_{\text{ref}}}(s_t, a_t) - V^{\pi_{\text{ref}}}(s_t)) \right] \\ &= J(\pi_{\text{ref}}) + \sum_{t=0}^{\infty} \sum_s \Pr\{s_t = s \mid \pi\} \sum_a \pi(s, a) \gamma^t (Q^{\pi_{\text{ref}}}(s, a) - V(s)) \\ &= J(\pi_{\text{ref}}) + \sum_s \sum_{t=0}^{\infty} \gamma^t \Pr\{s_t = s \mid \pi\} \sum_a \pi(s, a) (Q^{\pi_{\text{ref}}}(s, a) - V(s)) \\ &= J(\pi_{\text{ref}}) + \sum_s \eta_{\gamma}^{\pi}(s) \sum_a \pi(s, a) (Q^{\pi_{\text{ref}}}(s, a) - V(s)) \end{aligned}$$

Policy gradient theorem discounted case

Theorem (policy gradient for discounted episodic tasks)

For any differentiable stochastic policy π_{θ} and any MDP in the start-state setting it holds

$$\nabla_{\theta} J(\pi_{\theta}) = \sum_s \eta_{\gamma}^{\pi_{\theta}}(s) \sum_a \nabla_{\theta} \pi(s, a) Q^{\pi_{\theta}}(s, a) \ .$$

Policy gradient theorem discounted case, MC friendly

Corollary

For any differentiable stochastic policy π_{θ} and any MDP in the start-state setting it holds

$$\nabla_{\theta} J(\pi_{\theta}) \propto \mathbb{E}^{\pi_{\theta}} \left[\sum_{k=0}^{\infty} \gamma^k \frac{1}{\pi_{\theta}(s_k, a_k)} \nabla_{\theta} \pi(s_k, a_k) Q^{\pi_{\theta}}(s_k, a_k) \right] .$$

Policy gradient theorem discounted case, MC friendly, proof

Definition of improper state distribution and exchange of summation:

$$\nabla_{\theta} J(\pi_{\theta}) = \sum_{t=0}^{\infty} \gamma^t \sum_s \Pr\{s_t = s \mid \pi_{\theta}\} \sum_a \nabla_{\theta} \pi(s, a) Q^{\pi_{\theta}}(s, a)$$

Given trajectory $s_0, a_0, r_1, \dots, r_T$ following π_{θ} :

- states s_t occur with $\Pr\{s_t = s \mid \pi_{\theta}\}$
- weighting with γ^t is missing
- action a_k in the sequence was selected with probability $\pi(s_k, a_k)$, not uniformly, importance weighting with $\frac{1}{\pi(s_k, a_k)}$ ignores the numerator, thus only proportional

Deterministic policy gradient theorem, discounted

Corollary (deterministic policy gradient for discounted episodic tasks)

For any MDP with discount factor γ in the start-state setting and any deterministic policy π_{θ} it holds

$$\begin{aligned}\nabla_{\theta} J(\pi_{\theta}) &= \sum_s \eta_{\gamma}^{\pi_{\theta}}(s) \nabla_{\theta} \pi_{\theta}(s) \nabla_{\theta} Q^{\pi_{\theta}}(s, a) \Big|_{a=\pi_{\theta}(s)} \\ &\propto \mathbb{E}^{\pi_{\theta}} \left[\sum_{k=0}^{\infty} \gamma^k \nabla_{\theta} \pi_{\theta}(s) \nabla_{\theta} Q^{\pi_{\theta}}(s, a) \Big|_{a=\pi_{\theta}(s)} \right] \\ &= \mathbb{E}^{\pi_{\theta}} \left[\sum_{k=0}^{\infty} \gamma^k \nabla_{\theta} Q^{\pi}(s, \pi_{\theta}(s)) \right] .\end{aligned}$$

No $1/\pi_{\theta}(s, a)$ term as probability of action taken by deterministic policy always one.

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REINFORCE algorithm

- If s is sampled from distribution following π_{θ} and $\gamma = 1$, then

$$\sum_a \nabla_{\theta} \pi(s, a) Q^{\pi}(s, a)$$

is an unbiased estimate of $\nabla_{\theta} J(\pi_{\theta})$ – note equal weighting of actions (uniform distribution)

- If s and a are sampled from distribution following π_{θ} , then

$$\frac{1}{\pi(s, a)}$$

re-weights the samples ($\rightarrow$ importance weighting)

- $Q^{\pi}(s, a)$ is usually unknown; $Q^{\pi}(s_t, a_t)$ can be estimated using actual returns $R_t = \sum_{k=1}^{\infty} \gamma^{k-1} r_{t+k}$ or $R_t = \sum_{k=1}^{T-t} \gamma^{k-1} r_{t+k}$ (for a finite episode of length T) as estimates

REINFORCE pseudocode

$$\Delta \theta_t \propto \gamma^t \frac{1}{\pi(s_t, a_t)} \nabla_{\theta} \pi(s_t, a_t) R_t \stackrel{\text{"log derivative trick"}}{=} \gamma^t \underbrace{\nabla_{\theta} \ln \pi(s_t, a_t)}_{\text{"score function"}} R_t$$

Algorithm: REINFORCE

Input: differential policy π parameterized by θ , learning rate $\alpha > 0$, initial policy parameters θ

```

1 repeat
2   Generate episode  $s_0, a_0, r_1, \dots, s_{T-1}, a_{T-1}, r_T$ 
3   foreach  $t = 1, \dots, T - 1$  do
4      $R_t = \sum_{k=1}^{T-t} \gamma^{k-1} r_{t+k}$ 
5      $\theta \leftarrow \theta + \alpha R_t \gamma^t \nabla_{\theta} \ln \pi(s_t, a_t)$ 
6 until stopping criterion is met
```

Adding a baseline

The policy gradient theorem can be generalized to include a baseline, e.g.,

$$\nabla_{\boldsymbol{\theta}} J(\pi_{\boldsymbol{\theta}}) = \sum_s \eta_{\gamma}^{\pi_{\boldsymbol{\theta}}}(s) \sum_a \nabla_{\boldsymbol{\theta}} \pi(s, a) (Q^{\pi_{\boldsymbol{\theta}}}(s, a) - b(s)) \quad ,$$

where $b(s) : S \rightarrow \mathbb{R}$ is an arbitrary baseline function.

$\phi(s) = \sum_a \nabla_{\boldsymbol{\theta}} \pi(s, a) b(s)$ acts as a control variate. Note $\mathbb{E}[\phi] = 0$.

A possible choice of $b(s)$ would be some estimate of the value function $V^{\pi}(s)$. We call

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$$

the *advantage function*.

Policy gradients in the wild

In the wild, you will find several expressions for the policy gradient, which have the form

$$\mathbf{g} = \mathbb{E} \left[\sum_{t=0}^{\infty} \Psi_t \nabla_{\theta} \log \pi_{\theta}(s_t, a_t) \right] ,$$

where Ψ_t may be one of the following (Schulman et al., 2016):

- | | |
|---|--|
| 1. $\sum_{t=0}^{\infty} r_t$: total reward of the trajectory | 4. $Q^{\pi}(s_t, a_t)$: state-action value function |
| 2. $\sum_{t'=t}^{\infty} r_{t'}$: reward following action a_t | 5. $A^{\pi}(s_t, a_t)$: advantage function |
| 3. $\sum_{t'=t}^{\infty} r_{t'} - b(s_t)$: baselined version of previous formula | 6. $r_t + V^{\pi}(s_{t+1}) - V^{\pi}(s_t)$: temporal difference (TD) residual |

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